16/9/24

EXPERIMENT - 9

Implementation of Subnetting in CISCO PACKET TRACER simulator

AIM: To implement subnetting in cisco packet tracer by creating multiple subnets, assigning static IP address

Theory:

Subnetting is the process of dividing one network into smaller network called subnet.

It helps in efficient use of IP address and allow multiple network to be created from a single network address.

Requirements:

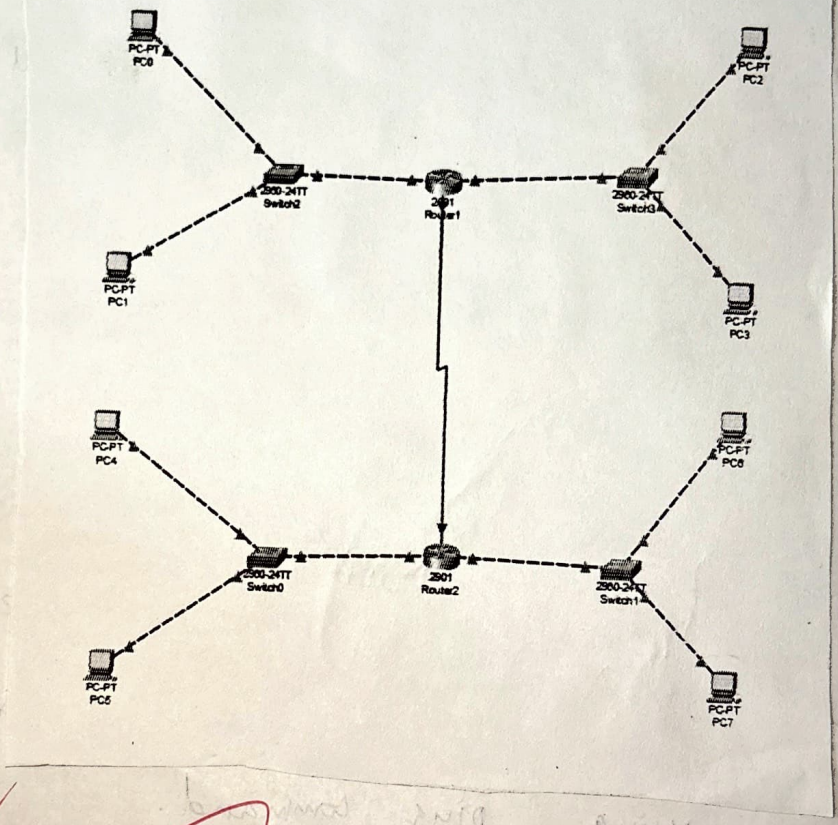
- Cisco packet tracer
- Router
- Switches
- PCs
- Copper straight-through cables

Procedure:

1. Open cisco packet tracer and create a new network topology.
2. Add the required router, switches and PCs to the workspace.
3. Connect the PCs to switch to the router using suitable cables.
4. Divide the network 192.168.1.0/24 units.
5. Assign static IP address to all PC and configure gateway.
6. Configure the router interface through the CLI
7. Configure IP address & subnet mask and default gateway on each PC
8. Check the connection b/w pcs using ping command.

9. Observe packet transmission with and without router.

Device	IP add	Subnet mask.	Gateway
Router	192.168.1.1	255.255.255.224	-
PC1	192.168.1.11	255.255.255.224	192.168.1.1
PC2	192.168.1.12	255.255.255.224	192.168.1.1
PC3	192.168.1.13	255.255.225.224	192.168.1.1



RESULT:

Thus, subnetting was successfully implemented in cisco packet tracer.